

READING PASSAGE 2

Big Data and Predictive Analytics

A In recent years, big data and predictive analytics has attracted considerable attention from both the academic community and the general public. This growing interest reflects a broader recognition of the importance of this topic for understanding some of the most pressing challenges facing contemporary society.

B Historically, scholarly understanding of this topic was limited by both theoretical frameworks and available methodologies. Early researchers operated within relatively narrow conceptual boundaries, and their findings, while groundbreaking for their time, often failed to capture the full complexity of the phenomena they sought to explain. The development of new research tools and analytical techniques in the latter half of the twentieth century fundamentally transformed the landscape, enabling scholars to investigate questions that had previously been considered beyond the reach of empirical inquiry.

C At the core of current understanding lies a set of interrelated mechanisms that collectively explain the observed patterns and outcomes. First, structural factors create the underlying conditions within which individual and collective action takes place. These structural dimensions include economic incentives, institutional frameworks, and cultural norms that shape behaviour in ways that are often not immediately apparent to the actors themselves. Second, dynamic feedback processes amplify or dampen initial effects, generating patterns of change that can diverge substantially from what linear models would predict.

D A substantial body of empirical evidence supports these theoretical frameworks. A landmark study conducted by researchers at leading international institutions analysed data spanning a thirty-year period and involving participants from more than forty countries. The findings revealed a consistent pattern of results that held across diverse cultural and economic contexts, suggesting that the identified mechanisms operate with considerable universality. The magnitude of the observed effects was particularly striking, with the key variables accounting for a substantial proportion of the total variance in outcomes.

E Several unresolved debates continue to animate scholarly discussion in this field. Perhaps the most prominent concerns the relative importance of structural versus agentic factors in determining outcomes. While some researchers emphasise the constraining power of institutional and economic structures, others highlight the capacity of individuals and groups to exercise agency and effect change even within seemingly restrictive contexts. This debate has significant practical implications, as it shapes recommendations for policy interventions and institutional reforms.

F The implications of this body of research extend well beyond the academy. As societies grapple with increasingly complex challenges, the insights generated by scholarly inquiry in this domain have the potential to inform more effective and equitable policy responses. Realising this potential, however, will require sustained investment in both fundamental research and the translational infrastructure needed to convert knowledge into action.

G In conclusion, the study of big data and predictive analytics represents a vital and dynamic area of contemporary research. The progress that has been made in understanding the key mechanisms and dynamics provides a solid foundation for both further scholarly inquiry and practical application. The challenges that remain should be seen not as reasons for discouragement but as opportunities for continued intellectual growth and social contribution.

Questions 1–2

Complete the sentences below. Choose NO MORE THAN THREE WORDS from the passage for each answer.

- 1 According to the passage, structural factors include economic incentives, institutional frameworks, and _____.
- 2 The author notes that researchers have called for more _____ approaches to investigation.

Question 3

Choose the correct letter, A, B, C or D.

- 3 The author suggests that early studies in this field were limited by:
A. minimal impact
B. creating underlying conditions for action
C. determining individual choices directly
D. irrelevant to outcomes

Questions 4–5

Do the following statements agree with the information given in the passage?

Write TRUE if the statement agrees with the information

Write FALSE if the statement contradicts the information

Write NOT GIVEN if there is no information on this

- 4 The landmark study discussed in the passage involved participants from multiple countries.
- 5 All critics agree that the research findings are universally applicable.

Question 6

Choose the correct letter, A, B, C or D.

- 6 What finding emerged from the landmark study discussed in the passage?
A. weak effects across studies
B. moderate to large effect sizes with consistency
C. contradictory findings
D. results only in specific populations

Question 7

Complete the sentences below. Choose NO MORE THAN THREE WORDS from the passage for each answer.

- 7 The passage identifies power _____ as a key factor at the organisational level.

Question 8

Choose the correct letter, A, B, C or D.

8 According to the passage, what is one of the key factors identified by researchers?

- A. excessive research funding
- B. translating academic findings into practical policy
- C. lack of public interest
- D. too many international collaborations

Question 9

Do the following statements agree with the information given in the passage?

Write TRUE if the statement agrees with the information

Write FALSE if the statement contradicts the information

Write NOT GIVEN if there is no information on this

9 Contemporary understanding of this topic is more sophisticated than it was fifty years ago.

Question 10

Choose the correct letter, A, B, C or D.

10 The passage indicates that critics have raised concerns about:

- A. inconsistent results across cultures
- B. a consistent pattern across diverse contexts
- C. no measurable effects
- D. effects limited to Western countries

Question 11

Complete the sentences below. Choose NO MORE THAN THREE WORDS from the passage for each answer.

11 Dynamic _____ processes can amplify or dampen initial effects in complex systems.

Question 12

Choose the correct letter, A, B, C or D.

12 What finding emerged from the landmark study discussed in the passage?

- A. pessimism about the future
- B. cautious optimism with acknowledgment of challenges
- C. complete certainty about outcomes
- D. indifference to practical applications

Question 13

Do the following statements agree with the information given in the passage?

Write TRUE if the statement agrees with the information

Write FALSE if the statement contradicts the information

Write NOT GIVEN if there is no information on this

13 All critics agree that the research findings are universally applicable.

